

Fake News Detection

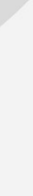
Explainable system using linguistic behavior and Explainable AI (LIME)



Introduction

Explainable Fake News Detection Using linguistic features to identify deceptive content and provide interpretable predictions.

Automated detection addresses rapid spread of misinformation and reduces manual verification overhead. The project emphasizes *explainability* and *trustworthiness* while maintaining realistic performance.





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Problem & Goals

Challenges

Existing models operate as black boxes, limiting stakeholder trust and adoption.

Manual verification is slow; models often overfit to dataset-specific cues. Need to balance **accuracy** with robust generalization across diverse real-world writing styles.





Transparency Needs

Provide interpretable decisions using methods like **LIME** and **SHAP** to expose word-level influences.

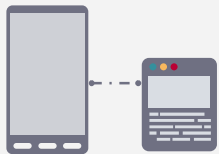
Explanations should clarify model limitations, highlight reliance on linguistic patterns, and enable human oversight for responsible deployment.



Project Objectives

Develop an automated, *explainable fake news detector* combining linguistic behavior and ML models to improve trust.

Extract features like sentiment, subjectivity, emotional intensity, and sensational language; integrate **LIME** for interpretability and evaluate on real-world data to ensure **robustness**.



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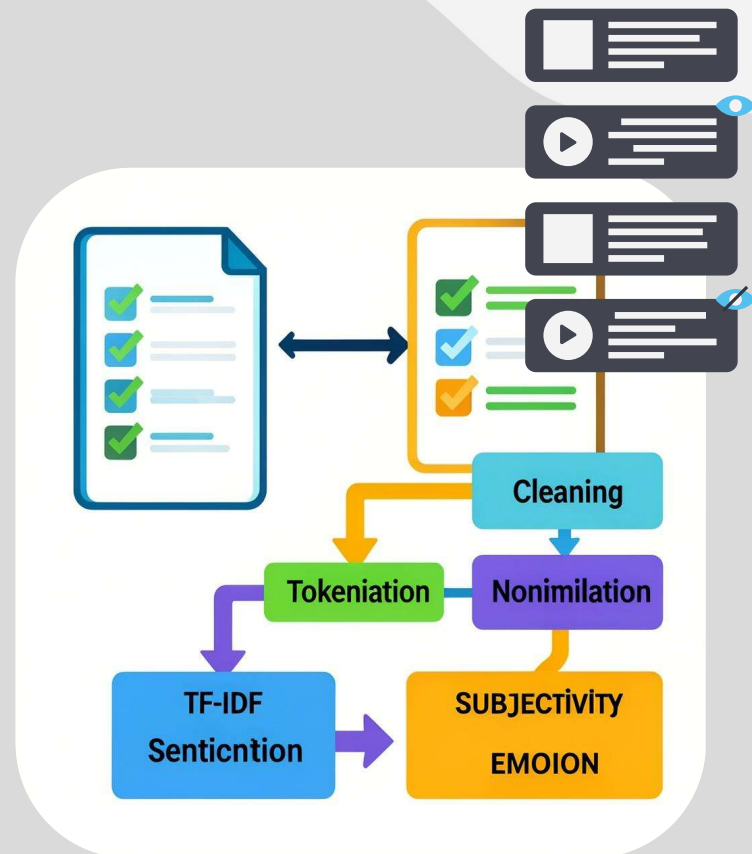
Method & Results



Data & Features

Used the *Fake Real News* dataset (Kaggle) with preprocessing: cleaning, tokenization, normalization.

Extract TF-IDF and linguistic features (sentiment, subjectivity, emotion, sensational markers) to capture writing style and tone for model inputs.



Models & Explainability

Train Logistic Regression, SVM, and Random Forest with confidence scores.

Apply *LIME* to produce word-level explanations, revealing which linguistic cues drive predictions and enabling human review of model decisions.

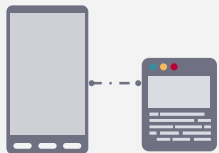
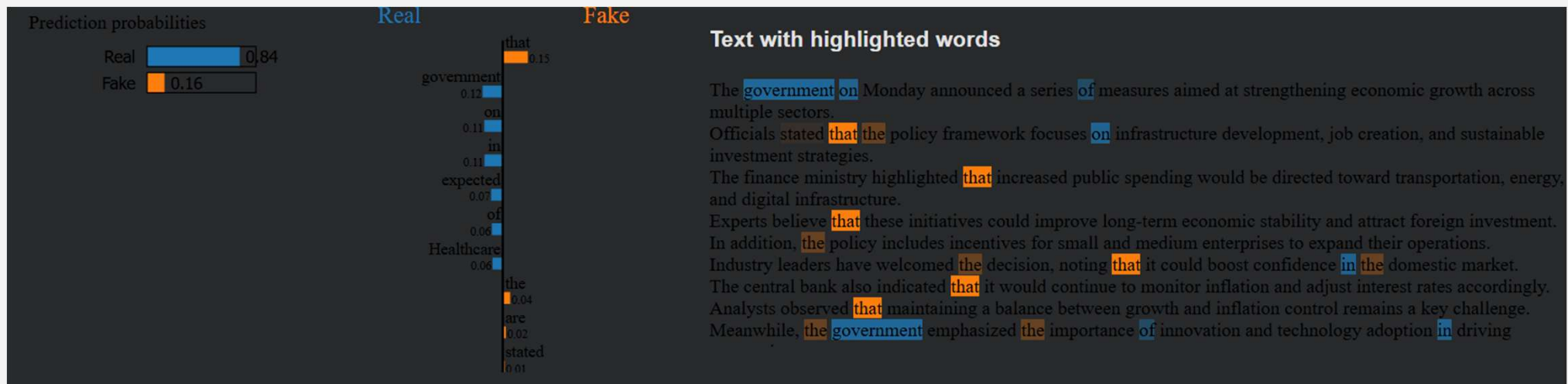


Evaluation & Future



Achieved ~84% dataset accuracy; evaluated using accuracy, precision, recall, and F1-score.

Observed reliance on linguistic patterns; emotional or sensational language signals fake claims. Future work: broader datasets, multimodal detection, real-time systems, and integrated fact-checking.



Conclusions



Developed an explainable detection system balancing **accuracy** and transparency using linguistic features.

Demonstrated realistic performance with identified generalization limits; explanations improve trust and enable targeted improvements for deployment in real-world settings.

